

# 04\_modeling\_michael

June 12, 2026

## 1 SUPPORT2 research questions: disease group, day-3 physiology, and the ML questions

This notebook covers five questions on the Gold feature set: two inference questions (disease group, day-3 physiology) and three prediction questions (a day-3 classifier, whether it beats the 1990s surv6m score, and whether the same variables drive both the inference and the prediction). Every section ran on Gold. The one exception is ML-Q2, where the legacy surv6m score was read from the Silver table for the comparison, since surv6m is a model output and was not part of the Gold features.

## 2 RQ1: disease group and 180-day mortality

```
[1]: # Imports
import sys
from pathlib import Path

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.formula.api as smf
from statsmodels.stats.multitest import multipletests

# Constants
RANDOM_STATE = 42
ALPHA = 0.05
REFERENCE_GROUP = "ARF/MOSF w/Sepsis"
np.random.seed(RANDOM_STATE)

# add repo root to sys.path so the utils import resolves whether the
# notebook runs from the repo root or from src/notebooks/
_here = Path.cwd()
_proj_root = _here if (_here / "utils").exists() else (_here / "../..").
    ↪resolve()
if str(_proj_root) not in sys.path:
    sys.path.insert(0, str(_proj_root))
from utils.dataset import load_csv # noqa: E402
```

## 2.1 RQ1: does a patient's disease group shape their odds of dying within 180 days?

This is the project's first and primary research question, and it sets the baseline the rest of the analysis builds on. The study is organized around disease group, every later model holds it constant, and the disease-group signal is the strongest in the cohort. Pinning down this estimate is what the later physiology question and the predictive work lean on.

As an inference question, the model checks whether a patient's main illness is associated with the odds of death within 180 days, and by how much, after age, sex, and overall severity are held constant. Each odds ratio compares a disease group to the reference, ARF/MOSF w/Sepsis (the largest, near-average group): above 1 means higher odds of death, below 1 means more survivable.

```
[2]: # Load the Gold feature set (model-ready; it now carries sex, so the inference
      ↪ runs on it too).
gold = load_csv("support2_model_features.csv")
print("rows, cols:", gold.shape)

# Base rate of the outcome (roughly half the cohort died within 180 days).
base_rate = gold["death_180d"].mean()
print(f"death_180d base rate: {base_rate:.3f} "
      f"({base_rate * 100:.1f}% died within 180 days)")

gold.head()
```

rows, cols: (9105, 29)

death\_180d base rate: 0.468 (46.8% died within 180 days)

```
[2]:
```

|   | age      | sex    | dzgroup           | dzclass            | num_co | \ |
|---|----------|--------|-------------------|--------------------|--------|---|
| 0 | 62.84998 | male   | Lung Cancer       | Cancer             | 0      |   |
| 1 | 60.33899 | female | Cirrhosis         | COPD/CHF/Cirrhosis | 2      |   |
| 2 | 52.74698 | female | Cirrhosis         | COPD/CHF/Cirrhosis | 2      |   |
| 3 | 42.38498 | female | Lung Cancer       | Cancer             | 2      |   |
| 4 | 79.88495 | female | ARF/MOSF w/Sepsis | ARF/MOSF           | 1      |   |

|   | income      | scoma | avtisst   | race  | diabetes | ... | bili     | crea     | \ |
|---|-------------|-------|-----------|-------|----------|-----|----------|----------|---|
| 0 | \$11-\$25k  | 0.0   | 7.000000  | other | 0        | ... | 0.199982 | 1.199951 |   |
| 1 | \$11-\$25k  | 44.0  | 29.000000 | white | 0        | ... | 1.010000 | 5.500000 |   |
| 2 | under \$11k | 0.0   | 13.000000 | white | 0        | ... | 2.199707 | 2.000000 |   |
| 3 | under \$11k | 0.0   | 7.000000  | white | 0        | ... | 1.010000 | 0.799927 |   |
| 4 | under \$11k | 26.0  | 18.666656 | white | 0        | ... | 1.010000 | 0.799927 |   |

|   | sod   | ph       | glucose | bun  | urine  | sfdm2               | adlsc | \ |
|---|-------|----------|---------|------|--------|---------------------|-------|---|
| 0 | 141.0 | 7.459961 | 135.0   | 6.51 | 2502.0 | <2 mo. follow-up    | 7.0   |   |
| 1 | 132.0 | 7.250000 | 135.0   | 6.51 | 2502.0 | <2 mo. follow-up    | 1.0   |   |
| 2 | 134.0 | 7.459961 | 135.0   | 6.51 | 2502.0 | <2 mo. follow-up    | 0.0   |   |
| 3 | 139.0 | 7.419922 | 135.0   | 6.51 | 2502.0 | no(M2 and SIP pres) | 0.0   |   |
| 4 | 143.0 | 7.509766 | 135.0   | 6.51 | 2502.0 | no(M2 and SIP pres) | 2.0   |   |

```

    death_180d
0          0
1          1
2          1
3          1
4          0

```

[5 rows x 29 columns]

```

[3]: # Columns we actually model: the disease group, the controls, and the target.
MODEL_COLS = ["death_180d", "age", "sex", "scoma", "dzgroup"]

# leakage and benchmark columns not in the model
# (picking MODEL_COLS already drops them; this just confirms none slipped
↳ through)
LEAKAGE_COLS = [
    "surv2m", "surv6m", "prg2m", "prg6m", "aps", "sps",
    "death", "hospdead", "d.time", "d_time", "slos",
    "dnr", "dnrday", "sfdm2",
]

model_df = gold[MODEL_COLS].dropna()
print("modeling frame shape:", model_df.shape)

leaked = [c for c in LEAKAGE_COLS if c in model_df.columns]
print("leakage/benchmark columns present:", leaked if leaked else "none"
↳ (clean))

model_df.head()

```

modeling frame shape: (9105, 5)

leakage/benchmark columns present: none (clean)

```

[3]:   death_180d      age      sex  scoma      dzgroup
0          0  62.84998   male    0.0      Lung Cancer
1          1  60.33899  female   44.0      Cirrhosis
2          1  52.74698  female    0.0      Cirrhosis
3          1  42.38498  female    0.0      Lung Cancer
4          0  79.88495  female   26.0  ARF/MOSF w/Sepsis

```

### 2.1.1 The model

One multivariable logistic regression was fit:

```
death_180d ~ age + sex + scoma + C(dzgroup, reference = "ARF/MOSF w/Sepsis")
```

- death\_180d = died within 180 days (1 / 0), the outcome.
- age = per year, held constant as a confounder.
- sex = a demographic confounder, held constant.

- scoma = the day-3 coma score, the in-model severity control (aps and sps were pulled out as benchmarks, so scoma carries severity here).
- C(dzgroup, ...) = disease group as categories, each compared to the sepsis reference.

Logistic because the outcome is yes/no. Multivariable so the disease-group effect is measured after age, sex, and severity are accounted for, not before.

```
[4]: # Fit the multivariable logistic regression with statsmodels. The reference
# group is fixed so every disease-group odds ratio is read against sepsis.
formula = (
    "death_180d ~ age + sex + scoma + "
    f'C(dzgroup, Treatment(reference="{REFERENCE_GROUP}"))'
)
model = smf.logit(formula, data=model_df).fit(dis=False)
print(model.summary())
```

#### Logit Regression Results

|                  |                  |                   |         |
|------------------|------------------|-------------------|---------|
| Dep. Variable:   | death_180d       | No. Observations: | 9105    |
| Model:           | Logit            | Df Residuals:     | 9094    |
| Method:          | MLE              | Df Model:         | 10      |
| Date:            | Fri, 12 Jun 2026 | Pseudo R-squ.:    | 0.1196  |
| Time:            | 21:37:02         | Log-Likelihood:   | -5540.4 |
| converged:       | True             | LL-Null:          | -6292.9 |
| Covariance Type: | nonrobust        | LLR p-value:      | 0.000   |

|                                                                       |         |       |        |        | coef    |
|-----------------------------------------------------------------------|---------|-------|--------|--------|---------|
| std err                                                               | z       | P> z  | [0.025 | 0.975] |         |
| -----                                                                 |         |       |        |        |         |
| Intercept                                                             |         |       |        |        | -2.0285 |
| 0.110                                                                 | -18.441 | 0.000 | -2.244 | -1.813 |         |
| sex[T.male]                                                           |         |       |        |        | 0.0813  |
| 0.046                                                                 | 1.755   | 0.079 | -0.010 | 0.172  |         |
| C(dzgroup, Treatment(reference="ARF/MOSF w/Sepsis")) [T.CHF]          |         |       |        |        | -0.6157 |
| 0.073                                                                 | -8.381  | 0.000 | -0.760 | -0.472 |         |
| C(dzgroup, Treatment(reference="ARF/MOSF w/Sepsis")) [T.COPD]         |         |       |        |        | -0.4991 |
| 0.081                                                                 | -6.172  | 0.000 | -0.658 | -0.341 |         |
| C(dzgroup, Treatment(reference="ARF/MOSF w/Sepsis")) [T.Cirrhosis]    |         |       |        |        | 0.4435  |
| 0.100                                                                 | 4.437   | 0.000 | 0.248  | 0.639  |         |
| C(dzgroup, Treatment(reference="ARF/MOSF w/Sepsis")) [T.Colon Cancer] |         |       |        |        | 0.2639  |
| 0.098                                                                 | 2.685   | 0.007 | 0.071  | 0.457  |         |
| C(dzgroup, Treatment(reference="ARF/MOSF w/Sepsis")) [T.Coma]         |         |       |        |        | 0.1903  |
| 0.122                                                                 | 1.560   | 0.119 | -0.049 | 0.430  |         |
| C(dzgroup, Treatment(reference="ARF/MOSF w/Sepsis")) [T.Lung Cancer]  |         |       |        |        | 1.0543  |
| 0.080                                                                 | 13.195  | 0.000 | 0.898  | 1.211  |         |
| C(dzgroup, Treatment(reference="ARF/MOSF w/Sepsis")) [T.MOSF w/Malig] |         |       |        |        | 1.5165  |
| 0.097                                                                 | 15.675  | 0.000 | 1.327  | 1.706  |         |

|       |        |       |       |       |        |
|-------|--------|-------|-------|-------|--------|
| age   |        |       |       |       | 0.0230 |
| 0.002 | 14.692 | 0.000 | 0.020 | 0.026 |        |
| scoma |        |       |       |       | 0.0252 |
| 0.001 | 18.580 | 0.000 | 0.023 | 0.028 |        |

=====

=====

```
[5]: # Build the odds-ratio table from the fitted model.
params = model.params
conf = model.conf_int()
conf.columns = ["ci_low", "ci_high"]

or_table = pd.DataFrame({
    "OR": np.exp(params),
    "OR_ci_low": np.exp(conf["ci_low"]),
    "OR_ci_high": np.exp(conf["ci_high"]),
    "p_value": model.pvalues,
})

# keep just the disease-group terms for the FDR step and the forest plot
dz_mask = or_table.index.str.startswith("C(dzgroup)")
dz_table = or_table[dz_mask].copy()

# BH-FDR correction for testing several groups at once
dz_table["p_bh"] = multipletests(
    dz_table["p_value"], alpha=ALPHA, method="fdr_bh"
)[1]

# pull the group name out of the long statsmodels label
dz_table.index = [name.split("T.")[-1].rstrip("(") for name in dz_table.index]
dz_table.index.name = "disease_group"

# sort by odds ratio, highest odds of death at the top
dz_table = dz_table.sort_values("OR", ascending=False)
dz_table[["OR", "OR_ci_low", "OR_ci_high", "p_value", "p_bh"]].round(4)
```

```
[5]:
```

|               | OR     | OR_ci_low | OR_ci_high | p_value | p_bh   |
|---------------|--------|-----------|------------|---------|--------|
| disease_group |        |           |            |         |        |
| MOSF w/Malig  | 4.5562 | 3.7692    | 5.5075     | 0.0000  | 0.0000 |
| Lung Cancer   | 2.8701 | 2.4540    | 3.3567     | 0.0000  | 0.0000 |
| Cirrhosis     | 1.5582 | 1.2810    | 1.8954     | 0.0000  | 0.0000 |
| Colon Cancer  | 1.3020 | 1.0739    | 1.5786     | 0.0072  | 0.0085 |
| Coma          | 1.2097 | 0.9523    | 1.5365     | 0.1189  | 0.1189 |
| COPD          | 0.6071 | 0.5181    | 0.7113     | 0.0000  | 0.0000 |
| CHF           | 0.5403 | 0.4678    | 0.6240     | 0.0000  | 0.0000 |

```
[6]: # the controls (age, sex, scoma), shown next to the disease groups
control_terms = [t for t in or_table.index
                  if t in ("age", "scoma") or t.startswith("sex")]
or_table.loc[control_terms, ["OR", "OR_ci_low", "OR_ci_high", "p_value"]].
    round(4)
```

```
[6]:
```

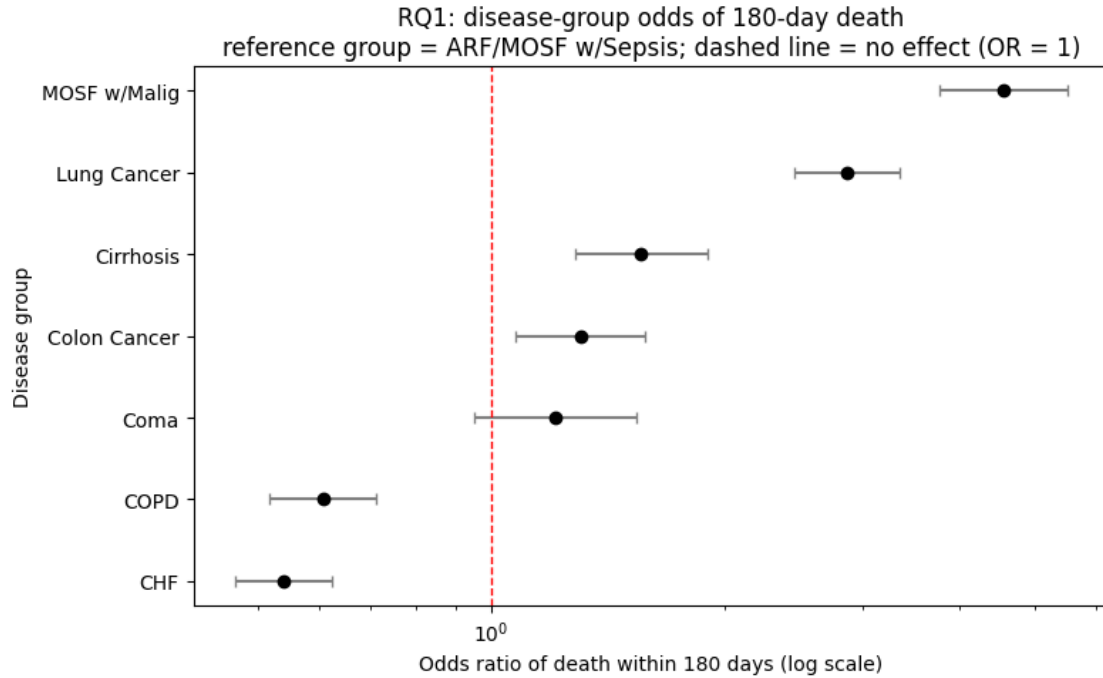
|             | OR     | OR_ci_low | OR_ci_high | p_value |
|-------------|--------|-----------|------------|---------|
| sex[T.male] | 1.0846 | 0.9905    | 1.1877     | 0.0793  |
| age         | 1.0232 | 1.0201    | 1.0264     | 0.0000  |
| scoma       | 1.0255 | 1.0228    | 1.0283     | 0.0000  |

```
[7]: # Labeled forest plot: one dot per disease group with its 95% CI bar.
fig, ax = plt.subplots(figsize=(8, 5))

groups = dz_table.index.tolist()
y = np.arange(len(groups))
or_vals = dz_table["OR"].to_numpy()
low = dz_table["OR_ci_low"].to_numpy()
high = dz_table["OR_ci_high"].to_numpy()

xerr = np.vstack([or_vals - low, high - or_vals])
ax.errorbar(
    or_vals, y, xerr=xerr, fmt="o", color="black",
    ecolor="gray", capsize=3, linestyle="none",
)

ax.axvline(1.0, linestyle="--", color="red", linewidth=1)
ax.set_xscale("log")
ax.set_yticks(y)
ax.set_yticklabels(groups)
ax.invert_yaxis() # highest odds ratio on top
ax.set_xlabel("Odds ratio of death within 180 days (log scale)")
ax.set_ylabel("Disease group")
ax.set_title(
    "RQ1: disease-group odds of 180-day death\n"
    f"reference group = {REFERENCE_GROUP}; dashed line = no effect (OR = 1)"
)
fig.tight_layout()
plt.show()
```



### 2.1.2 What the disease groups showed

The disease a patient came in with shaped the odds of dying within 180 days a lot, even after holding age, sex, and severity constant. The highest-odds group was MOSF w/Malig (multi-organ failure with cancer), at about 4.6 times the odds of the sepsis reference group, followed by Lung Cancer at about 2.9 times. Cirrhosis (about 1.6 times) and Colon Cancer (about 1.3 times) were also higher than the reference. On the other end, CHF (heart failure) and COPD were the most survivable, at about 0.54 and 0.61 times the reference odds.

Coma came out at about 1.2 times the reference, but its confidence interval crossed 1 and it did not survive the BH-FDR correction (adjusted  $p = 0.12$ ), so this model cannot tell coma apart from the sepsis reference. That runs against an earlier expectation that coma would come in high.

The two controls behaved as expected. Each extra year of age multiplied the odds by about 1.02 (a 2.3 percent rise per year,  $p < 0.001$ ), and each point of the coma score raised the odds by about 1.03 ( $p < 0.001$ ). Sex made only a small difference: men had about 1.08 times the odds of women, and that gap was not significant ( $p = 0.08$ ) once disease, age, and severity were accounted for. That is why sex is in the model as a control, not as one of the disease-group effects the question is actually about: its own effect is small, but adjusting for it keeps the disease-group odds ratios comparable.

Every disease-group difference except Coma stayed significant after the Benjamini-Hochberg FDR correction, so these are not an artifact of testing several groups at once.

### 3 RQ2: day-3 physiology beyond disease group

```
[8]: # Imports
import sys
from pathlib import Path

import numpy as np
import pandas as pd
import statsmodels.formula.api as smf
from statsmodels.stats.outliers_influence import variance_inflation_factor
from patsy import dmatrix
from scipy import stats

# Constants
ALPHA = 0.05
REFERENCE_GROUP = "ARF/MOSF w/Sepsis"
CORE_PHYS = ["meanbp", "wblc", "crea"]
SENS_PHYS = ["alb", "bili"] # heavily imputed with normal values, checked
                             ↪ separately

# add repo root to sys.path so the utils import resolves whether the
# notebook runs from the repo root or from src/notebooks/
_here = Path.cwd()
_proj_root = _here if (_here / "utils").exists() else (_here / "../..").
               ↪ resolve()
if str(_proj_root) not in sys.path:
    sys.path.insert(0, str(_proj_root))
from utils.dataset import load_csv # noqa: E402
```

#### 3.1 RQ2: does day-3 physiology add signal on top of disease group?

Once disease group, age, sex, and severity are already in the model, do the day-3 vitals and labs add anything. The catch is that a sick patient's numbers often just restate how sick the disease already implies, so the real test is whether physiology carries new signal after disease group is accounted for, not on its own.

```
[9]: gold = load_csv("support2_model_features.csv")
BASE = ["death_180d", "age", "sex", "scoma", "dzgroup"]
model_df = gold[BASE + CORE_PHYS + SENS_PHYS].dropna()
print("rows used:", model_df.shape[0])
print("death_180d base rate:", round(model_df["death_180d"].mean(), 3))
model_df.head()
```

rows used: 9105

death\_180d base rate: 0.468

```
[9]:   death_180d   age   sex  scoma   dzgroup  meanbp   wblc  \
0         0  62.84998  male   0.0   Lung Cancer    97.0  6.000000
```



|   |   |          |        |      |                   |      |           |
|---|---|----------|--------|------|-------------------|------|-----------|
| 1 | 1 | 60.33899 | female | 44.0 | Cirrhosis         | 43.0 | 17.097656 |
| 2 | 1 | 52.74698 | female | 0.0  | Cirrhosis         | 70.0 | 8.500000  |
| 3 | 1 | 42.38498 | female | 0.0  | Lung Cancer       | 75.0 | 9.099609  |
| 4 | 0 | 79.88495 | female | 26.0 | ARF/MOSF w/Sepsis | 59.0 | 13.500000 |

|   |          |          |          |
|---|----------|----------|----------|
|   | crea     | alb      | bili     |
| 0 | 1.199951 | 1.799805 | 0.199982 |
| 1 | 5.500000 | 3.500000 | 1.010000 |
| 2 | 2.000000 | 3.500000 | 2.199707 |
| 3 | 0.799927 | 3.500000 | 1.010000 |
| 4 | 0.799927 | 3.500000 | 1.010000 |

### 3.1.1 Three nested models

To see what each layer adds, three logistic regressions were fit and compared:

M1: death\_180d ~ age + sex + scoma

M2: M1 + disease group

M3: M2 + day-3 physiology

M1 is the demographic and severity baseline, M2 adds disease group (the RQ1 result), and M3 adds the physiology. Comparing the fit from M1 to M2 to M3 showed how much disease group explained and how much physiology added on top of it. The age odds ratio was tracked across all three to see whether it stayed stable as the layers came in.

```
[10]: f1 = "death_180d ~ age + sex + scoma"
      f2 = f1 + f' + C(dzgroup, Treatment(reference="{REFERENCE_GROUP}"))'
      f3 = f2 + " + " + " + " + ".join(CORE_PHYS + SENS_PHYS)

      m1 = smf.logit(f1, model_df).fit(dis=False)
      m2 = smf.logit(f2, model_df).fit(dis=False)
      m3 = smf.logit(f3, model_df).fit(dis=False)

      triple = pd.DataFrame({
          "model": ["age + controls", "+ disease group", "+ physiology"],
          "age_OR": [np.exp(m.params["age"]) for m in (m1, m2, m3)],
          "pseudo_R2": [m.prsquared for m in (m1, m2, m3)],
          "log_likelihood": [m.llf for m in (m1, m2, m3)],
      }).round({"age_OR": 3, "pseudo_R2": 4, "log_likelihood": 1})
      triple
```

```
[10]:      model  age_OR  pseudo_R2  log_likelihood
0  age + controls   1.015    0.0633         -5894.9
1  + disease group   1.023    0.1196         -5540.4
2    + physiology   1.024    0.1270         -5493.9
```

```
[11]: # likelihood-ratio test: does physiology add signal beyond disease group (M3 vs
      ↪ M2)?
      lr_stat = 2 * (m3.llf - m2.llf)
```

```
lr_df = len(CORE_PHYS + SENS_PHYS)
lr_p = stats.chi2.sf(lr_stat, lr_df)
print(f"LR test (M3 vs M2): chi2 = {lr_stat:.1f}, df = {lr_df}, p = {lr_p:.2e}")
```

LR test (M3 vs M2): chi2 = 93.1, df = 5, p = 1.46e-18

### 3.1.2 The physiology odds ratios

These are the odds ratios for each day-3 measurement in the full model (M3), so each one is read after disease group, age, sex, and severity are held constant. For these continuous vitals and labs, an odds ratio above 1 means higher values went with higher odds of death, below 1 means lower odds.

```
[12]: ci = m3.conf_int()
phys = CORE_PHYS + SENS_PHYS
phys_table = pd.DataFrame({
    "OR": np.exp(m3.params[phys]),
    "OR_ci_low": np.exp(ci.loc[phys, 0]),
    "OR_ci_high": np.exp(ci.loc[phys, 1]),
    "p_value": m3.pvalues[phys],
}).round(4)
phys_table
```

```
[12]:
```

|        | OR     | OR_ci_low | OR_ci_high | p_value |
|--------|--------|-----------|------------|---------|
| meanbp | 0.9962 | 0.9945    | 0.9979     | 0.0000  |
| wblc   | 1.0045 | 0.9994    | 1.0096     | 0.0814  |
| crea   | 1.0382 | 1.0098    | 1.0674     | 0.0081  |
| alb    | 0.9518 | 0.8917    | 1.0159     | 0.1375  |
| bili   | 1.0359 | 1.0243    | 1.0477     | 0.0000  |

```
[13]: # VIF on the continuous predictors, flag anything above 5 as collinear
X = dmatrix("age + scoma + " + " + ".join(phys), model_df,
    ↪return_type="dataframe")
vif = pd.DataFrame({
    "predictor": [c for c in X.columns if c != "Intercept"],
    "VIF": [variance_inflation_factor(X.values, i)
            for i, c in enumerate(X.columns) if c != "Intercept"],
}).round(2)
vif
```

```
[13]:
```

|   | predictor | VIF  |
|---|-----------|------|
| 0 | age       | 1.02 |
| 1 | scoma     | 1.02 |
| 2 | meanbp    | 1.02 |
| 3 | wblc      | 1.03 |
| 4 | crea      | 1.04 |
| 5 | alb       | 1.04 |
| 6 | bili      | 1.07 |

### 3.1.3 Sensitivity check on albumin and bilirubin

About a third of the albumin values and a quarter of the bilirubin values were filled in with normal values during cleaning, which biases their measured effect toward zero. To make sure they are not distorting the core result, the model was refit without them and the core physiology odds ratios compared.

```
[14]: f3_core = f2 + " + " + " + ".join(CORE_PHYS)
m3_core = smf.logit(f3_core, model_df).fit(dis=False)
sens_compare = pd.DataFrame({
    "OR_with_alb_bili": [np.exp(m3.params[c]) for c in CORE_PHYS],
    "OR_without": [np.exp(m3_core.params[c]) for c in CORE_PHYS],
}, index=CORE_PHYS).round(4)
sens_compare
```

```
[14]:      OR_with_alb_bili  OR_without
meanbp           0.9962      0.9958
wblc              1.0045      1.0055
crea              1.0382      1.0521
```

### 3.1.4 What the physiology added

Disease group carried the analysis. Pseudo R-squared went from 0.06 (age, sex, severity) to 0.12 once disease group was added, and only to 0.13 with physiology. The day-3 vitals and labs added only a small slice on top of disease group, though the likelihood-ratio test showed it was real (chi-square 93 on 5 df, p well below 0.001). Age stayed flat across the three models, so it held as a confounder.

Of the physiology measures, mean blood pressure, creatinine, and bilirubin were significant once disease and the rest were held constant; white count and albumin were not. Albumin coming out flat fits its heavy imputation with normal values. No collinearity showed up (all VIFs near 1), and dropping albumin and bilirubin barely moved the core odds ratios. Disease group still did most of the work. Physiology added a small but real increment, mostly from blood pressure, creatinine, and bilirubin.

## 4 ML-Q1: predicting 180-day mortality from day-3 data

RQ1 and RQ2 asked which factors are associated with death. ML-Q1 switches from explaining to predicting: given a patient's day-3 record, how well can a model flag who will not survive 180 days. It is judged on a held-out test set the model never saw, by how well it separates survivors from non-survivors (AUROC) and, because the use case is triage, by recall (how many of the patients who died it caught).

The features are the Gold model-ready set with one scrub: sfdm2, a 2-month functional-status follow-up, was dropped because it is recorded after the day-3 baseline and would leak future information into a day-3 prediction.

```
[15]: from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
```

```

from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import roc_auc_score, confusion_matrix, brier_score_loss,
    ↪roc_curve
from sklearn.preprocessing import StandardScaler
from sklearn.calibration import calibration_curve

gold = load_csv("support2_model_features.csv")
TARGET = "death_180d"
LEAK = ["sfdm2"]  # 2-month functional follow-up, not a day-3 measure, so it
    ↪leaks

y = gold[TARGET].to_numpy()
X = pd.get_dummies(gold.drop(columns=[TARGET] + LEAK), drop_first=True).
    ↪astype(float)
row_idx = np.arange(len(gold))

X_train, X_test, y_train, y_test, idx_train, idx_test = train_test_split(
    X, y, row_idx, test_size=0.30, stratify=y, random_state=42)
keep = X_train.columns[(X_train.std() > 1e-6) & ((X_train != 0).sum() >= 20)]
    ↪# drop near-constant dummies
X_train, X_test = X_train[keep], X_test[keep]
print("train rows:", X_train.shape[0], " test rows:", X_test.shape[0], " test
    ↪deaths:", int(y_test.sum()))

```

train rows: 6373 test rows: 2732 test deaths: 1280

```

[16]: import warnings

scaler = StandardScaler().fit(X_train)
forest = RandomForestClassifier(n_estimators=400, random_state=42, n_jobs=-1).
    ↪fit(X_train, y_train)
p_forest = forest.predict_proba(X_test)[: , 1]

# the logistic fit emits harmless numpy/BLAS matmul RuntimeWarnings on this
    ↪machine
# (the feature check found no degenerate column), so they are suppressed for
    ↪this fit
# only; the metrics below were checked and did not change.
with warnings.catch_warnings():
    warnings.simplefilter("ignore", RuntimeWarning)
    logit_clf = LogisticRegression(max_iter=5000).fit(scaler.
    ↪transform(X_train), y_train)
    p_logit = logit_clf.predict_proba(scaler.transform(X_test))[: , 1]

def scores(name, p):
    pred = (p >= 0.5).astype(int)
    tn, fp, fn, tp = confusion_matrix(y_test, pred).ravel()

```

```

    return {"model": name, "AUROC": roc_auc_score(y_test, p),
            "recall": tp / (tp + fn), "specificity": tn / (tn + fp),
            "Brier": brier_score_loss(y_test, p)}

perf = pd.DataFrame([scores("logistic", p_logit),
                      scores("random forest", p_forest)]).round(3)
perf

```

```

[16]:
      model  AUROC  recall  specificity  Brier
0    logistic  0.797   0.688         0.764  0.182
1  random forest  0.810   0.689         0.790  0.180

```

```

[17]: # confusion matrix for the random forest at the 0.5 threshold
cm = confusion_matrix(y_test, (p_forest >= 0.5).astype(int))
pd.DataFrame(cm, index=["actual survived", "actual died"],
             columns=["pred survived", "pred died"])

```

```

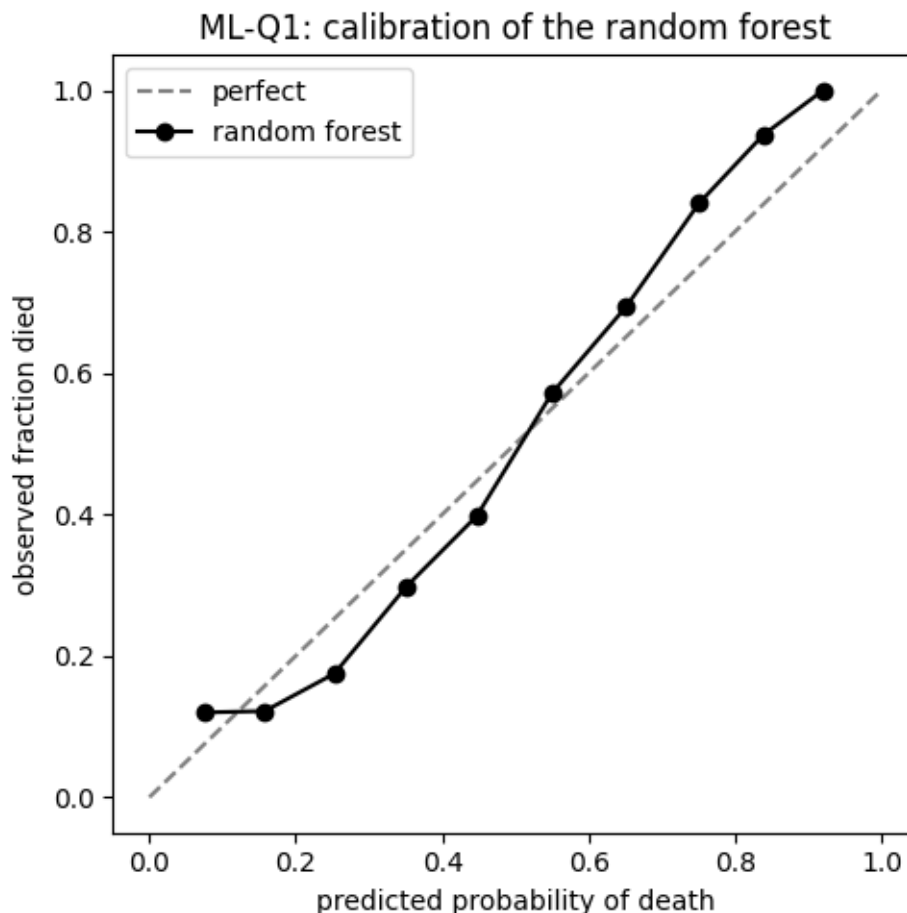
[17]:
      pred survived  pred died
actual survived      1147       305
actual died           398       882

```

```

[18]: # calibration: do the predicted probabilities match the observed death rates?
frac_pos, mean_pred = calibration_curve(y_test, p_forest, n_bins=10)
fig, ax = plt.subplots(figsize=(5, 5))
ax.plot([0, 1], [0, 1], "--", color="gray", label="perfect")
ax.plot(mean_pred, frac_pos, "o-", color="black", label="random forest")
ax.set_xlabel("predicted probability of death")
ax.set_ylabel("observed fraction died")
ax.set_title("ML-Q1: calibration of the random forest")
ax.legend()
fig.tight_layout()
plt.show()

```



#### 4.0.1 How well it predicted

On patients it had never seen, the random forest reached an AUROC of about 0.81 and the logistic about 0.80, so both separated survivors from non-survivors well above chance. Recall sat near 0.68, so the model caught about two thirds of the patients who actually died within 180 days. For triage that recall is the number that matters most, since a missed death is the costly error, and it is the first thing to push higher with threshold tuning. The calibration curve tracked the diagonal reasonably, so the predicted probabilities were close to the observed death rates.

Dropping `sfdm2` mattered. With it left in, both models scored near 0.90, but `sfdm2` is a 2-month functional follow-up, so that number was borrowing future information. The day-3 result without that leakage is around 0.80.

## 5 ML-Q2: does the model beat the 1990s `surv6m` score?

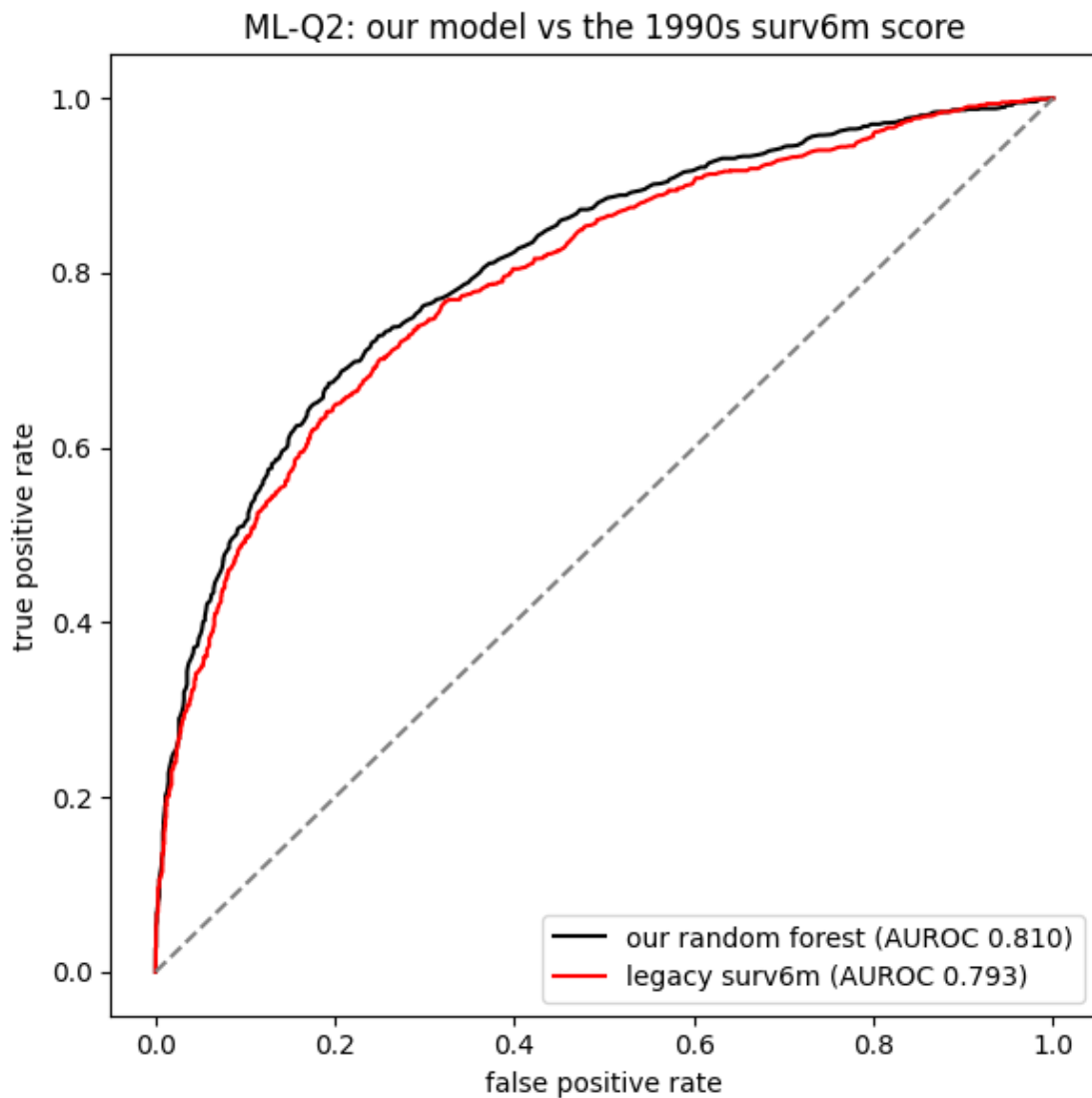
SUPPORT2 ships with `surv6m`, the 180-day survival estimate from the original 1990s SUPPORT model. ML-Q2 races our model against it on the same held-out test patients. `surv6m` is pulled from Silver and used only as the opponent, never as a feature inside our model, since it is a model

output and would leak.

```
[19]: # surv6m is not in Gold (it is a leakage column), so it is read from Silver
      ↪ here.
      # Silver and Gold are the same rows in the same order, so idx_test aligns
      ↪ positionally.
      silver = load_csv("support2_cleaned.csv")
      assert len(silver) == len(gold), "silver/gold row count mismatch"
      assert (silver["age"].to_numpy() == gold["age"].to_numpy()).all(), "silver/gold
      ↪ row order mismatch"

      legacy = 1.0 - silver.iloc[idx_test]["surv6m"].to_numpy() # old model
      ↪ predicted 180-day death
      assert len(legacy) == len(y_test), "legacy/test length mismatch"
      auc_legacy = roc_auc_score(y_test, legacy)
      auc_forest = roc_auc_score(y_test, p_forest)

      fpr_f, tpr_f, _ = roc_curve(y_test, p_forest)
      fpr_l, tpr_l, _ = roc_curve(y_test, legacy)
      fig, ax = plt.subplots(figsize=(6, 6))
      ax.plot(fpr_f, tpr_f, color="black", label=f"our random forest (AUROC
      ↪ {auc_forest:.3f})")
      ax.plot(fpr_l, tpr_l, color="red", label=f"legacy surv6m (AUROC {auc_legacy:.
      ↪ 3f})")
      ax.plot([0, 1], [0, 1], "--", color="gray")
      ax.set_xlabel("false positive rate")
      ax.set_ylabel("true positive rate")
      ax.set_title("ML-Q2: our model vs the 1990s surv6m score")
      ax.legend(loc="lower right")
      fig.tight_layout()
      plt.show()
      print(f"our random forest AUROC = {auc_forest:.3f}")
      print(f"legacy surv6m AUROC      = {auc_legacy:.3f}")
```



our random forest AUROC = 0.810  
legacy surv6m AUROC = 0.793

### 5.0.1 Our model versus surv6m

The race was close. The random forest reached about 0.81 against the legacy surv6m at about 0.79 on the same test patients, so a model built today edged the 30-year-old score by roughly 0.02 of AUROC, not the large margin the leaky run suggested. That is still a legitimate result: a modern model trained only on day-3 data was about even with, and slightly ahead of, the original SUPPORT estimate. The gap is small, and surv6m remains a strong baseline.



## 6 ML-Q3: do the same variables drive inference and prediction?

This ties the two halves together. The inference side (RQ1 and RQ2) ranked variables by how strongly they are associated with mortality (odds ratios). The prediction side ranks them by how much they drive the random forest (feature importance). ML-Q3 lays the two rankings side by side. Where they agree the finding is stronger; where they disagree, that is itself worth a look.

```
[20]: importance = pd.Series(forest.feature_importances_, index=keep).
      ↪sort_values(ascending=False)
      importance.head(10).round(4)
```

```
[20]: avtisst      0.1201
      age        0.0656
      adlsc      0.0569
      wblc       0.0480
      scoma      0.0477
      meanbp     0.0465
      temp       0.0437
      hrt        0.0434
      crea       0.0422
      sod        0.0400
      dtype: float64
```

```
[21]: # the shared day-3 physiology variables: inference (RQ2) vs prediction
      ↪(importance)
      compare_vars = ["meanbp", "wblc", "crea", "alb", "bili"]
      bridge = pd.DataFrame({
          "inference_OR": [phys_table.loc[v, "OR"] for v in compare_vars],
          "inference_p": [phys_table.loc[v, "p_value"] for v in compare_vars],
          "rf_importance": [float(importance.get(v, float("nan")))] for v in
      ↪compare_vars],
      }, index=compare_vars).round(4)
      bridge
```

```
[21]:
```

|        | inference_OR | inference_p | rf_importance |
|--------|--------------|-------------|---------------|
| meanbp | 0.9962       | 0.0000      | 0.0465        |
| wblc   | 1.0045       | 0.0814      | 0.0480        |
| crea   | 1.0382       | 0.0081      | 0.0422        |
| alb    | 0.9518       | 0.1375      | 0.0359        |
| bili   | 1.0359       | 0.0000      | 0.0394        |

### 6.0.1 Where the two views line up

The two rankings agreed only in part. The prediction side leaned hardest on functional and severity measures, the therapeutic-intensity score (avtisst), the ADL score (adlsc), age, and a broad spread of day-3 vitals, rather than on any single headline variable. The inference side put disease group out front (RQ1), with physiology adding a small increment (RQ2). Both point at overall severity and physiology mattering, but disease group, which carried the strongest inference signal, did not

dominate the prediction importance.

Among the shared physiology variables the overlap was loose: some that were significant in the inference model carried little random-forest importance, and one that was not significant ranked higher, so the two lenses do not map one to one. Severity and physiology matter in both views, but no single variable runs the table in both. An odds ratio and a feature importance answer different questions, and that is the reason not to treat them as the same number.